Data Cleaning and Pre-processing Report

India Pension Scheme Dataset

This report documents the data-quality assessment, cleaning, validation, transformation, and preprocessing procedures applied before exploratory data analysis. The explanation is based on the latest project notebook and the cleaning decisions documented during the project.

1. Purpose of Data Cleaning

The purpose of cleaning was to improve the reliability and consistency of the dataset before statistical analysis and visualization. The project specifically examined missing values, duplicate records, inconsistent formats, incorrect data types, logically impossible values, and unusual financial observations. The cleaning process deliberately distinguished between values that were clearly invalid and values that were extreme but potentially genuine.

The notebook first explored the structure and quality of the data before making changes. This included reviewing the first and last records, dataset dimensions, column names, data types, descriptive statistics, missing values, and categorical frequencies. The initial exploration identified issues such as missing values, inconsistent categorical representations, date-format differences, numerical fields stored as text, and logically invalid numerical values.

2. Initial Data Quality Assessment

- The dataset contained 38 columns covering demographic, employment, pension, financial, and other subscriber-related information.
- The initial data-type review showed a mixture of categorical/textual and numerical variables. Several quantitative fields were stored as object/text values even though they represented numbers.
- Missing observations were identified in variables including Date of Birth, Age, Marital Status, City, Employer Name, Annual Income, and Current Fund Value.
- Date variables such as Date of Birth, Date of Joining, and Last Transaction Date required conversion to consistent date formats.
- The column name PAN_Linked was identified as having a leading space, illustrating the need for column-name and formatting checks.
- Categorical variables contained inconsistent representations. For example, Gender included variations such as Male, MALE, male, and M, and corresponding variations for Female and Other.
- The exploratory stage also identified logically suspicious values such as negative ages, impossible ages, negative financial amounts, negative asset-allocation percentages, and other inconsistent values.

3. Date and Time-Related Cleaning

Date fields were standardized so that they could be reliably compared and used in later calculations. The cleaning process examined Date of Birth, Date of Joining, and Last Transaction Date.

- Date values were converted to a consistent date representation.
- Future or logically impossible Date of Birth values were identified and treated as invalid.
- Date information was used to validate Age rather than relying only on the original Age field.
- Date consistency checks helped distinguish genuine values from values caused by data-entry or formatting problems.

4. Age Validation and Correction

Age required special validation because it is directly related to Date of Birth and retirement-related variables. The notebook identified 26 negative age records. These were not all treated in the same way: the corresponding Date of Birth was examined to determine whether the negative sign represented a correctable sign error.

- Validated negative ages were corrected when the corresponding Date of Birth supported the correction.
- A negative age associated with an invalid future Date of Birth could not be reliably validated and was therefore treated separately.
- A negative age with missing Date of Birth could not be independently validated.
- Ages of 150, 210, and 999 years were classified as logically impossible and converted to missing values.
- After cleaning, Age was cross-checked against Date of Birth to confirm internal consistency.

This approach is preferable to simply deleting all unusual ages because it uses related information in the dataset to determine whether an apparent error can be corrected with evidence.

5. Numeric Data-Type Conversion

Several quantitative variables were initially stored as object/text data types. These fields were converted to numeric form before statistical analysis, correlation analysis, aggregation, and outlier detection.

- Annual_Income_INR
- Monthly_Contribution_INR

- Employer_Monthly_Contribution_INR
- Years_of_Contribution
- Total_Contribution_Till_Date_INR
- Current_NAV_INR
- Total_Units_Held
- Expected_Monthly_Pension_INR
- Relevant asset-allocation percentage variables

Numeric conversion used coercion for invalid non-numeric entries so that they could be identified as missing rather than causing the cleaning process to fail. This ensured that subsequent descriptive statistics and visualizations were performed on appropriately typed variables.

6. Financial and Contribution Validation

Financial variables were checked using logical rules because some values were impossible for the corresponding variables. The objective was to correct clearly invalid observations while preserving valid high-value observations.

- Negative Annual Income was treated as logically invalid.
- Extremely large Annual Income observations were investigated rather than automatically classified as errors.
- Negative Monthly Contribution values were treated as invalid.
- Negative Total Contribution values were treated as invalid.
- Negative Current NAV values were treated as invalid.
- Negative Total Units Held values were treated as invalid.
- Negative Expected Monthly Pension values were treated as invalid.
- Negative Years of Contribution values were treated as invalid because a contribution period cannot logically be negative.
- Zero Employer Monthly Contribution values were retained where they occurred as valid observations across different occupation groups; they were not automatically treated as missing.

7. Treatment of Extreme Financial Values

Extreme financial values were investigated separately from logically invalid values. This distinction was important because a very large pension contribution or fund value may be unusual but still genuine.

- Annual income contained a negative value and an extremely large maximum value. The negative value was treated as invalid, while the extreme high value was investigated in relation to the rest of the distribution.
- The analysis considered the mean, median, quartiles, and maximum when assessing extreme income observations.
- Clearly erroneous extreme values identified during the project were replaced using the median of valid observations where justified by the cleaning decision.
- The cleaning process did not automatically remove every statistical outlier. Instead, values were assessed for logical plausibility and context.

8. Asset Allocation Validation

Asset-allocation variables were checked against their logical percentage range. Equity, Corporate Debt, Government Securities, and Alternative Investment percentages should represent proportions of an investment allocation.

- Values below 0% were treated as invalid and converted to missing values.
- The logical 0%–100% range was used as the validation boundary.
- Potentially unusual but logically valid percentages were retained.
- This prevented impossible allocation values from distorting the later investment-choice and asset-allocation analyses.

9. Missing-Value Handling

Missing values were reviewed before and after cleaning. An important principle of the project was that missing values should not be filled automatically without a defensible basis.

- Values that were found to be logically invalid during cleaning were converted to missing values.
- Remaining genuine missing observations were retained when there was insufficient evidence to infer their correct values.
- The final missing-value review confirmed that missingness increased in some numeric variables because invalid values had correctly been converted to missing.
- The largest remaining missingness was observed in variables such as Annual Income, Monthly Contribution, Total Contribution Till Date, Age, and Years of Contribution.

- Categorical variables generally had fewer missing observations, with State and Pension Scheme Type having only one missing value each in the final review.
- No blanket imputation strategy was applied because different variables have different meanings and inappropriate imputation could introduce artificial information.

10. Categorical Data Quality

Categorical frequencies were examined to identify inconsistent labels, unusual categories, and missing-value representations. The exploration showed multiple representations of the same category, particularly in Gender.

- Gender contained capitalization and abbreviation differences such as Male/MALE/male/M, Female/FEMALE/female/F, and Other/other/OTHER/O.
- Such inconsistencies can produce incorrect counts and misleading group comparisons if left untreated.
- Categorical standardization was therefore considered necessary before frequency-based analysis and visualization.
- The exploration stage retained the original values so that the cleaning decisions could be documented separately from the initial data-quality assessment.

11. Duplicate Record Removal

An exact duplicate check identified 4 duplicate rows. Duplicate observations can cause double-counting and can distort descriptive statistics, group counts, and visualizations.

- The 4 exact duplicate records were identified.
- Duplicate rows were removed using drop_duplicates(keep='first').
- The first occurrence of each duplicated record was retained.
- The cleaned dataset was subsequently checked to confirm that duplicate records had been removed.

12. Outlier Detection and Treatment

Outlier analysis was performed after logically invalid values had been addressed. This distinction is important: a value can be statistically extreme without being incorrect.

- The Interquartile Range (IQR) method was used to identify potential statistical outliers.
- Boxplots were used to visually examine the spread and potential outliers of important financial variables.

- Variables such as Current Fund Value and Expected Monthly Pension showed substantial dispersion.
- High-value observations were investigated rather than automatically deleted.
- Potentially genuine high contributions, fund values, or pension amounts were retained because they may represent real differences among subscribers.
- This approach prevents legitimate high-value observations from being incorrectly removed simply because they are far from the centre of the distribution.

13. Internal Consistency Checks

Related variables were checked against each other to identify logical inconsistencies. One important check examined the relationship between Total Units Held, Current NAV, and Current Fund Value.

- Current Fund Value was checked against the relationship between Total Units Held and Current NAV.
- Contribution-related variables were reviewed for logical consistency.
- Age was validated against Date of Birth.
- These cross-variable checks provided additional evidence that the cleaned data could be used for subsequent analysis.

14. Transformations and Derived Variables Used for Analysis

After cleaning, the dataset was transformed into analysis-ready forms where required. The project used grouped and derived representations to make comparisons easier without changing the underlying cleaned observations.

- Contribution experience was grouped into meaningful ranges for comparison of fund values and expected pension.
- Annual income was divided into quartile-based income groups for comparative analysis.
- Monthly contribution was similarly grouped into contribution bands for financial outcome comparisons.
- Investment choices and pension scheme types were used as grouping variables for summary tables and comparative analysis.
- These derived groupings were used for analysis and visualization while retaining the underlying continuous financial variables.

15. Final Cleaning

The overall cleaning strategy followed a conservative and evidence-based approach. Values were changed only when there was a logical or analytical reason to do so. Clearly impossible values were corrected or converted to missing values, while potentially genuine extreme observations were retained.

- Logical invalidity was treated differently from statistical extremeness.
- Related variables were used to validate questionable records whenever possible.
- Missing values were not blindly imputed.
- Duplicates were explicitly identified and removed.
- Data types and date formats were standardized before analysis.
- Outliers were investigated rather than automatically deleted.
- Final consistency checks were performed before the cleaned data was used for EDA.

16. Outcome of the Cleaning Process

The result of the cleaning process was an analysis-ready dataset with standardized formats, appropriate numerical data types, invalid observations addressed, duplicate records removed, and potential outliers assessed. The cleaned dataset was then used for descriptive analysis, group-wise comparisons, correlation analysis, and visualization.

The cleaning process therefore provided the foundation for reliable exploratory analysis while minimizing the risk of introducing artificial values or removing legitimate observations.